Topic 3: Analog Modulation

CSE 4405: Data and Telecommunications

Md. Tanvir Hossain Saikat

April 28, 2026

Islamic University of Technology (IUT)

Learning Objectives

1. Explain why a low-frequency analog message is shifted onto a carrier
2. Distinguish clearly among **baseband**, **carrier**, and **passband**
3. Describe AM intuitively in time domain and spectrum domain
4. Explain sidebands, AM bandwidth, and why the carrier is power-inefficient
5. Compare standard AM, DSB-SC, SSB, and VSB without relying on heavy Fourier analysis
6. Explain FM and PM as **angle-modulation** methods
7. Relate analog modulation to practical systems and to later topics in this course

Reading focus: Forouzan 5e, Ch. 5, §5.2 (pp. 147–151); Rappaport 2e, Ch. 6 for modulation intuition. Goldsmith Ch. 5 is kept as a bridge to later digital modulation, not as the main source for AM/FM/PM definitions.

Start from the Problem, Not from the Formula

What we have

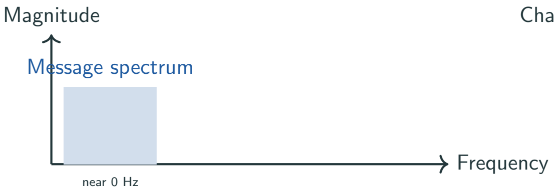
Voice, music, and many sensor outputs are **baseband** signals.

Their important frequencies are clustered near 0 Hz (slow variations)

What the channel gives us

Many practical channels are effectively **band-pass**.

They work well only around some nonzero frequency band.



So the goal is simple: move the message from baseband to a band centered around a carrier frequency f_c.

Why Not Radiate the Baseband Signal Directly?

- Efficient antennas are usually some fraction of the wavelength
- Wavelength is $\lambda = \frac{c}{f}$
- For a 3 kHz audio component, $\lambda \approx 100$ km
- A quarter-wave antenna would then be about 25 km long
- A 1 MHz carrier has $\lambda \approx 300$ m, so a quarter-wave antenna is about 75 m

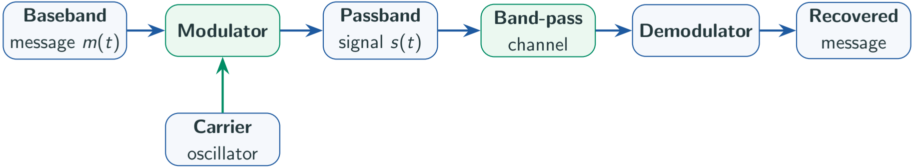
Engineering reason

Low-frequency messages are excellent for **information**, but poor for **radiation**. A higher-frequency carrier makes practical antennas and spectrum allocation possible.

Takeaway

Modulation is not decorative. It solves a physical problem: **how to place information where the medium and antenna can actually support transmission.**

Baseband to Passband: The Big Picture



Vocabulary

Modulation creates the passband signal.
Demodulation tries to recover the original message from the received carrier-based waveform.

The Carrier Gives Us Three Knobs

Carrier knob	What it means physically	Modulation family
Amplitude	How large the sinusoid is	AM
Frequency	How quickly the sinusoid oscillates	FM
Phase	Where the sinusoid is in its cycle relative to a reference	PM

Unmodulated carrier:

$$c(t) = A_c \cos(2\pi f_c t + \phi)$$

Analog modulation changes one of these carrier parameters according to the message $m(t)$.

Roadmap for the Rest of the Topic

1. **AM first:** easiest place to see how a message becomes a changing carrier and how sidebands appear
2. **AM variants next:** DSB-SC, SSB, and VSB are not new ideas; they are engineering responses to AM's power and bandwidth waste
3. **Angle modulation last:** FM and PM trade more bandwidth for better noise behavior
4. **End with context:** where these systems appear and why this topic matters before Topics 4 and 5

Why this order is natural

AM shows the basic mechanism most transparently. Once you see **carrier + sidebands + demodulation**, the later schemes feel like purposeful variations, not separate facts to memorize.

One General Form for Analog Modulation

All analog modulated signals can be written as:

$$s(t) = A(t)\cos[\theta(t)]$$

Amplitude and phase both can vary

Phase has carrier and modulation:

$$\theta(t) = 2\pi f_c t + \phi(t)$$

- If $A(t)$ varies with message $m(t) \rightarrow$ **Amplitude Modulation**
- If $\phi(t)$ varies with message $m(t) \rightarrow$ **FM or PM**

AM: Let the Envelope Follow the Message

Definition

In **Amplitude Modulation (AM)**, the carrier amplitude varies with the message, while the carrier frequency and phase remain fixed.

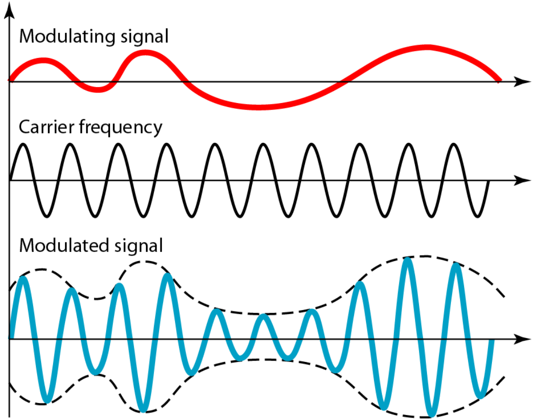
$$s_{\mathrm{AM}}(t) = A_c\left[1 + k_a m(t)\right]\cos(2\pi f_c t)$$

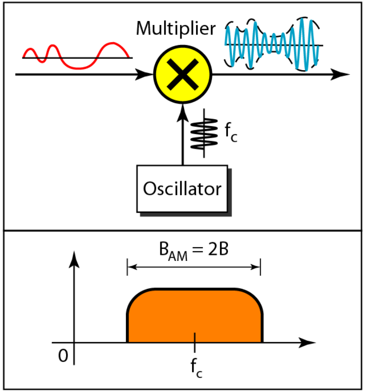
The factor in brackets controls the envelope.

What the receiver watches

If the modulation is done properly, the outer shape of the AM waveform closely matches the message. That outer shape is called the **envelope**.

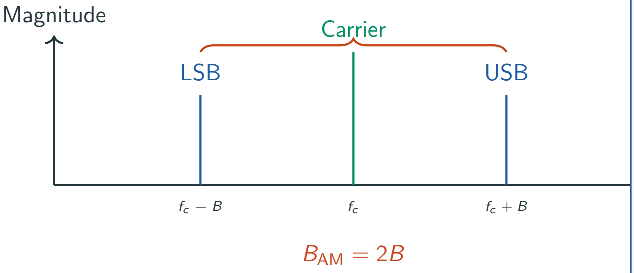
AM in the Time Domain





Fast carrier oscillation shaped by slow message envelope.

AM Bandwidth and Sidebands



Sideband intuition

If the message bandwidth is B, AM creates two shifted copies of that message spectrum:

- **Lower sideband (LSB)**
- **Upper sideband (USB)**

So the total occupied bandwidth doubles.

Bandwidth Rule: $B_{\text{AM}} = 2B$

For example: 5 kHz speech $\rightarrow$ 10 kHz bandwidth needed

Standard AM: General and Modulation Index

AM equation:

$$s_{\text{AM}}(t) = A_c[1 + k_a m(t)] \cos(2\pi f_c t)$$

Envelope $[1 + k_a m(t)]$ varies with message

For single-tone message $m(t) = A_m \cos(2\pi f_m t)$:

$$s_{\text{AM}}(t) = A_c[1 + k_a A_m \cos(2\pi f_m t)] \cos(2\pi f_c t)$$

Define modulation index:

$$\mu = k_a A_m$$

Controls envelope depth; must satisfy $0 < \mu \leq 1$

Standard AM: Expanded to Sidebands

Rewrite with modulation index:

$$s(t) = A_c \cos(2\pi f_c t) + A_c \mu \cos(2\pi f_m t) \cos(2\pi f_c t)$$

Carrier plus product term

Apply product-to-sum: $\cos A \cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$

$$s(t) = A_c \cos(2\pi f_c t) + \frac{A_c \mu}{2} \cos(2\pi (f_c + f_m) t) + \frac{A_c \mu}{2} \cos(2\pi (f_c - f_m) t)$$

Carrier + upper sideband + lower sideband

Standard AM: What Frequencies Are Present?

Expanded AM equation with single-tone message:

$$s(t) = A_c \cos(2\pi f_c t) + \frac{A_c \mu}{2} \cos(2\pi(f_c + f_m)t) + \frac{A_c \mu}{2} \cos(2\pi(f_c - f_m)t)$$

Carrier + upper sideband + lower sideband

Component	**Frequency**	**Role**
Carrier	f_c	Power only
Upper SB	$f_c + f_m$	Message info
Lower SB	$f_c - f_m$	Message info

Information is in the sidebands; carrier is wasted power.

What Information Is Actually Useful in AM?

Carrier

Large spectral line at f_c

Useful as a built-in reference for easy reception

But it does **not** carry new message content

Sidebands

Upper and lower sidebands contain the message information

They are shifted versions of the baseband spectrum

They are the parts we really need to preserve

Why later AM variants exist

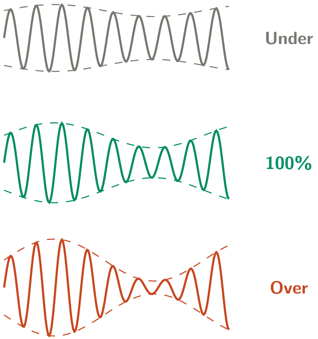
Once engineers realized that the carrier costs power and the two sidebands duplicate information, it became natural to ask: **can we remove some of that waste without losing the message?**

Modulation Depth: Too Little, Just Right, Too Much

For AM, the **modulation index** or **modulation depth** measures how strongly the message changes the carrier amplitude:

$$\mu = k_a m_p$$

Here m_p is the normalized peak message amplitude.



- $\mu < 1$: under-modulated, envelope is small but recoverable
- $\mu = 1$: 100% modulation, strong but still safe
- $\mu > 1$: over-modulated, envelope crosses itself and distorts

Detection rule: envelope detection needs a clean envelope; over-modulation destroys it.

Why Ordinary AM Was Attractive

Strengths

Simple transmitter concept
Simple receiver concept
Cheap broadcast receivers possible
Easy tuning for mass-market radio

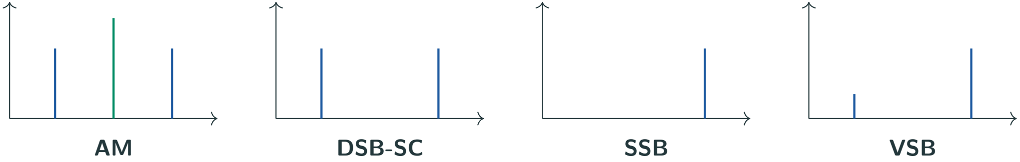
Weaknesses

Needs $2B$ bandwidth
Wastes power on the carrier
Sensitive to amplitude noise and fading
Poor efficiency compared with later variants

Historical reason

AM stayed important not because it was ideal, but because the total system was simple enough to deploy widely. Communication engineering often accepts inefficiency in one place to reduce cost somewhere else.

From AM to the AM Family



One design question drives all four schemes

How much of the carrier and how many sidebands should we keep? Different answers give different tradeoffs in **power efficiency**, **bandwidth efficiency**, and **receiver complexity**.

DSB-SC: Save Carrier Power

Idea

Double Sideband Suppressed Carrier (DSB-SC) keeps both sidebands but removes the large carrier component.

$$s_{\text{DSB-SC}}(t) = A_c m(t) \cos(2\pi f_c t)$$

The envelope is no longer a simple always-positive outline.

What improves

Power is not wasted on a carrier that carries no new information. Bandwidth stays the same as AM:

$$B_{\text{DSB-SC}} = 2B$$

Receiver implication: without a strong transmitted carrier, the receiver must recreate a matching carrier reference. Power efficiency improves, but demodulation becomes less simple.

DSB-SC: General Form

Suppress the carrier term in AM:

$$s_{\text{DSB-SC}}(t) = A_c m(t) \cos(2\pi f_c t)$$

Message directly multiplies the cosine; no $A_c[1 + \ldots]$ envelope

For a single-tone message $m(t) = A_m \cos(2\pi f_m t)$:

$$s_{\text{DSB-SC}}(t) = A_c A_m \cos(2\pi f_m t) \cos(2\pi f_c t)$$

Product of two cosines; apply product-to-sum formula

Power efficiency

No carrier power wasted; all power goes to sidebands

DSB-SC: Frequency Content

Apply product-to-sum formula $\cos A \cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$:

$$s_{\text{DSB-SC}}(t) = \frac{A_c A_m}{2} \cos(2\pi(f_c + f_m)t) + \frac{A_c A_m}{2} \cos(2\pi(f_c - f_m)t)$$

Two sidebands, no carrier

Component	**Frequency**
Upper sideband	$f_c + f_m$
Lower sideband	$f_c - f_m$

All transmission power carries message information.

SSB: Save Bandwidth Too

Idea

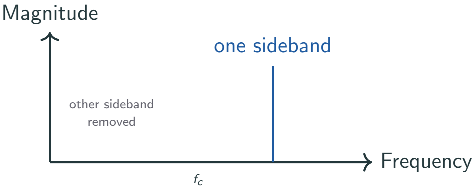
Single Sideband (SSB) sends only one sideband, either upper or lower, and usually suppresses the carrier as well.

Bandwidth win

Only one shifted copy of the message is transmitted:

$$B_{\text{SSB}} = B$$

So SSB uses roughly half the bandwidth of ordinary AM.



Why SSB matters: scarce spectrum makes one-sideband voice transmission attractive.

SSB Equation: Keep Only One Sideband

DSB-SC has both sidebands:

$$s(t) = \frac{A_c A_m}{2}\cos(2\pi(f_c + f_m)t) + \frac{A_c A_m}{2}\cos(2\pi(f_c - f_m)t)$$

Upper sideband + lower sideband

SSB keeps only one:

$$s_{\text{SSB}}(t) = \frac{A_c A_m}{2}\cos(2\pi(f_c \pm f_m)t)$$

Either $f_c + f_m$ (USB) or $f_c - f_m$ (LSB)

Bandwidth savings

Uses only B bandwidth instead of $2B$; half the spectrum space

How Filtering Separates USB and LSB

DSB-SC contains both sidebands:

$$s(t) = \frac{A_c A_m}{2} \cos(2\pi(f_c + f_m)t) + \frac{A_c A_m}{2} \cos(2\pi(f_c - f_m)t)$$

Upper sideband at $f_c + f_m$ and lower at $f_c - f_m$

Use bandpass filtering:

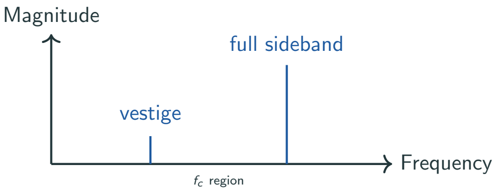
- **To get USB**: pass $f_c + f_m$, reject $f_c - f_m$
- **To get LSB**: pass $f_c - f_m$, reject $f_c + f_m$

$$\text{DSB-SC} \xrightarrow{\text{BPF}} \text{SSB (USB or LSB)}$$

VSB: A Practical Compromise

Idea

Vestigial Sideband (VSB) transmits one full sideband plus a small leftover piece, or **vestige**, of the other sideband.



- Not as bandwidth-hungry as full AM
- Not as filter-demanding as ideal SSB
- Chosen when theory says "remove one sideband" but hardware says "not perfectly"

Why VSB: Practical filters are imperfect; VSB balances filtering demands.

VSB Equation: Concept

VSB = **Vestigial Sideband**: transmits one full sideband plus a small "vestige" of the other.

Mathematical form:

$$S_{\text{VSB}}(f) = S_{\text{DSB-SC}}(f)H_{\text{VSB}}(f)$$

Take DSB-SC and apply VSB filter $H_{\text{VSB}}(f)$

Filter role

Filter $H_{\text{VSB}}(f)$ keeps one sideband strong and weakens the other

VSB vs SSB: Bandwidth Tradeoff

SSB Ideal

One sideband completely removed

$$B_{\text{SSB}} = B$$

Minimal bandwidth

VSB Practical

Small vestige remains

$$B < B_{\text{VSB}} < 2B$$

Practical compromise

Perfect SSB filtering is hard. VSB is engineering's practical answer.

AM Family Compared

Scheme	Carrier sent?	Bandwidth	Main benefit	Main cost
AM	Yes	$2B$	Very simple envelope detection	Poor power efficiency
DSB-SC	No	$2B$	Saves carrier power	Needs coherent detection
SSB	Usually no	B	Saves both power and bandwidth	Harder filtering and receiver design
VSB	Usually reduced	Between B and $2B$	Practical compromise between SSB and AM	Still more complex than AM

Pattern to notice

Every step away from plain AM improves efficiency, but each step asks more from the receiver. This **efficiency versus complexity tradeoff** appears throughout communication systems.

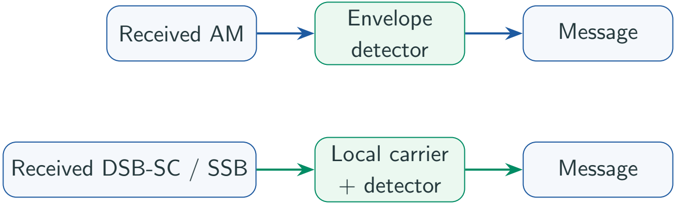
How the Message Is Recovered

Envelope detection for AM

If a strong carrier is transmitted and the waveform is not over-modulated, the receiver can follow the envelope directly. This is why broadcast AM receivers can be simple.

Coherent detection for DSB-SC / SSB

If the transmitted carrier is missing or reduced, the receiver must create a local carrier reference with matching frequency and phase before recovering the message.



Angle Modulation: Keep Amplitude Constant

What changes

In **angle modulation**, the information is carried by the **angle** of the sinusoid, meaning its frequency behavior or phase behavior, not by its amplitude.

Why this matters

If noise mainly shakes the amplitude, a receiver can often limit or ignore that amplitude variation before detection. That is why FM is famous for better noise behavior than AM.

$$s(t) = A_c \cos\bigl(2\pi f_c t + \phi(t)\bigr)$$

For FM and PM, the carrier amplitude A_c stays nearly constant while the angle term $\phi(t)$ carries the message.

FM: Let the Message Bend the Frequency

Definition

In **Frequency Modulation (FM)**, the **instantaneous frequency** of the carrier moves above or below f_c according to the message amplitude.

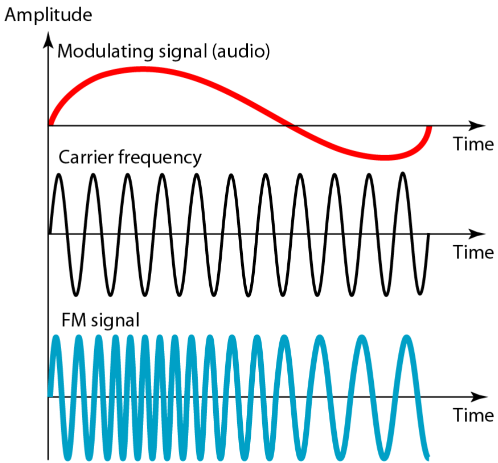
$$s_{\mathrm{FM}}(t) = A_c \cos\left(2\pi f_c t + 2\pi k_f \int m(t)\, dt\right)$$

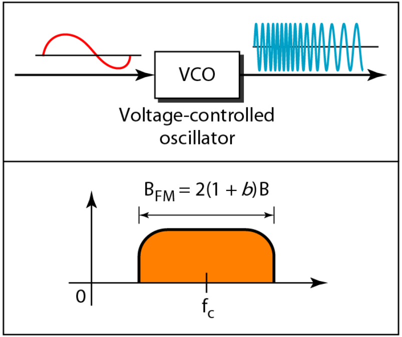
Do not focus on the integral. The physical meaning is the important part: **message amplitude controls frequency deviation.**

Reading the waveform

Where the FM waveform cycles are closer together, the instantaneous frequency is higher. Where they are farther apart, the instantaneous frequency is lower.

FM in the Time Domain





Key reading rule: As the amplitude of the information signal changes, the frequency of the carrier changes proportionately

FM: General Form

All angle-modulated signals:

$$s(t) = A_c \cos[\theta(t)]$$

Amplitude constant, phase $\theta(t)$ varies with message

For FM, the phase includes an integral of the message:

$$s_{\text{FM}}(t) = A_c \cos\left[2\pi f_c t + 2\pi k_f \int m(t)dt\right]$$

Integral means message amplitude directly controls frequency deviation

Key Insight

Phase $\theta(t) = 2\pi f_c t + 2\pi k_f \int m(t)dt$ integrates the message

FM: Instantaneous Frequency

Instantaneous frequency is the rate of phase change:

$$f_i(t) = \frac{1}{2\pi}\frac{d\theta(t)}{dt}$$

Derivative of phase gives instantaneous frequency

Differentiate FM phase with respect to time:

$$\frac{d\theta(t)}{dt} = 2\pi f_c + 2\pi k_f m(t)$$

Integral disappears; message remains

Therefore:

$$f_i(t) = f_c + k_f m(t)$$

Instantaneous frequency = Carrier frequency + Message-scaled deviation

Why Does FM Contain Integration?

The Design Goal: Make frequency follow the message amplitude

If we naively wrote:

$$\theta(t) = 2\pi f_c t + 2\pi k_f m(t)$$

we'd get a phase that varies with $m(t)$, not frequency varying with $m(t)$.

The Fix: Integrate the message first:

$$\theta(t) = 2\pi f_c t + 2\pi k_f \int m(t)dt$$

Why it works

When we differentiate to find frequency, $\frac{d}{dt} \int m(t)dt = m(t)$. So the integral in phase ensures frequency is proportional to message amplitude.

FM with Single-Tone: Setup and Integral

Let the message be a single tone:

$$m(t) = A_m \cos(2\pi f_m t)$$

Single frequency at f_m with amplitude A_m

Substitute into FM equation:

$$s_{\text{FM}}(t) = A_c \cos\left[2\pi f_c t + 2\pi k_f \int A_m \cos(2\pi f_m t) dt\right]$$

Integral of cosine is needed

Standard integral:

$$\int \cos(2\pi f_m t) dt = \frac{1}{2\pi f_m} \sin(2\pi f_m t)$$

FM with Single-Tone: Standard Form

After evaluating the integral:

$$s_{\mathrm{FM}}(t) = A_c \cos\left[2\pi f_c t + \frac{k_f A_m}{f_m} \sin(2\pi f_m t)\right]$$

Phase has sine function (integral of cosine)

Define FM modulation index:

$$\beta_f = \frac{\Delta f}{f_m} = \frac{k_f A_m}{f_m}$$

Ratio of frequency deviation to message frequency

Standard form:

$$s_{\mathrm{FM}}(t) = A_c \cos[2\pi f_c t + \beta_f \sin(2\pi f_m t)]$$

Carson's bandwidth: $B_{\mathrm{FM}} = 2(\Delta f + B) = 2(k_f A_m + f_m)$

Frequency Deviation and FM Bandwidth

Deviation

The maximum amount by which the instantaneous frequency moves away from f_c is the **frequency deviation** Δf. Larger message amplitude usually means larger deviation.

Forouzan gives an empirical FM bandwidth rule:

$$B_{\mathrm{FM}} \approx 2(1+\beta)B$$

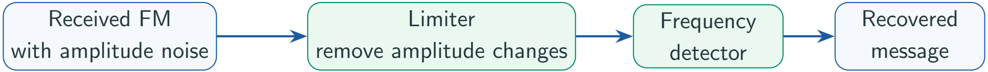
where B is message bandwidth and β is a modulation factor.

Broadcast FM

Stereo audio extends to roughly 15 kHz. Broadcast FM is allocated about 200 kHz per station, which shows that FM buys noise performance by spending more bandwidth.

Tradeoff: AM is bandwidth-cheap but noise-sensitive. FM is noise-robust but bandwidth-hungry. Better performance often costs more spectrum.

Why FM Handles Amplitude Noise Better



AM problem

If noise changes amplitude, AM suffers immediately because amplitude is where the message lives.

FM advantage

If noise changes amplitude, an FM receiver can often suppress much of that distortion before extracting the message, because the message lives in frequency changes.

PM: Let the Message Bend the Phase

Definition

In **Phase Modulation (PM)**, the message changes the carrier **phase** directly.

$$s_{\mathrm{PM}}(t) = A_c \cos\big(2\pi f_c t + k_p m(t)\big)$$

The carrier amplitude stays essentially constant.

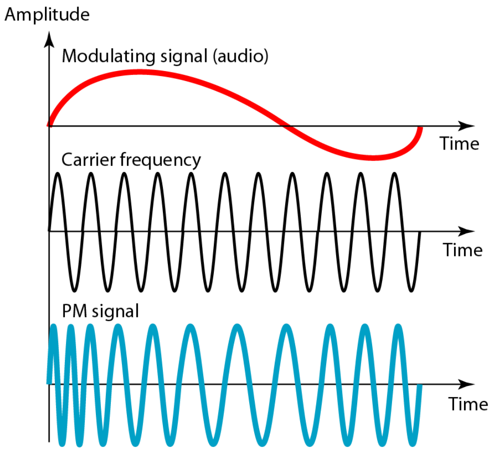
How to think about it

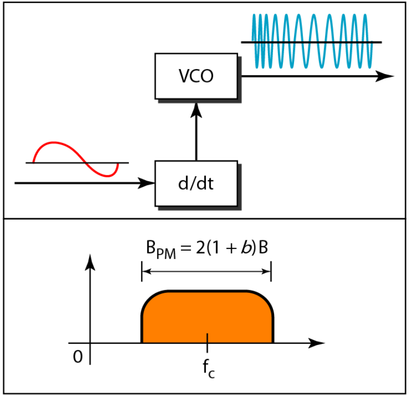
If the message changes slowly, the phase rotates slowly. If the message changes rapidly, phase changes rapidly, and that also creates instantaneous frequency changes.

Relationship to FM

FM and PM are close relatives. FM ties the instantaneous frequency to the message amplitude. PM ties the instantaneous phase to the message amplitude.

PM in the Time Domain





Key reading rule: The phase of the carrier signal is modulated to follow the changing voltage (amplitude) of the modulating signal

PM: General Form

For PM, the phase is directly proportional to the message:

$$s_{\text{PM}}(t) = A_c \cos[2\pi f_c t + k_p m(t)]$$

Phase changes with message value (not its integral)

The instantaneous phase:

$$\theta(t) = 2\pi f_c t + k_p m(t)$$

Direct coupling between message and phase

Key Difference from FM

Phase is $k_p m(t)$ (direct), not $2\pi k_f \int m(\tau)d\tau$ (integral)

PM: Instantaneous Frequency

Instantaneous frequency is the derivative of phase divided by 2π:

$$f_i(t) = \frac{1}{2\pi}\frac{d\theta(t)}{dt}$$

Rate of phase change gives frequency

Differentiate PM phase:

$$\frac{d\theta(t)}{dt} = 2\pi f_c + k_p \frac{dm(t)}{dt}$$

Message derivative appears in the frequency term

Therefore:

$$f_i(t) = f_c + \frac{k_p}{2\pi}\frac{dm(t)}{dt}$$

Instantaneous frequency depends on message slope, not message amplitude

PM with Single-Tone: Setup and Index

Let the message be a single tone:

$$m(t) = A_m \cos(2\pi f_m t)$$

Single frequency at f_m with amplitude A_m

Substitute into PM equation:

$$s_{\text{PM}}(t) = A_c \cos[2\pi f_c t + k_p A_m \cos(2\pi f_m t)]$$

Phase is modulated by a cosine function

Define the PM modulation index:

$$\beta_p = k_p A_m$$

Phase deviation index (analogous to FM's β_f)

PM with Single-Tone: Standard Form

Standard PM form for single-tone message:

$$s_{\mathrm{PM}}(t) = A_c \cos[2\pi f_c t + \beta_p \cos(2\pi f_m t)]$$

Phase deviation is proportional to message amplitude

Compare with FM single-tone form:

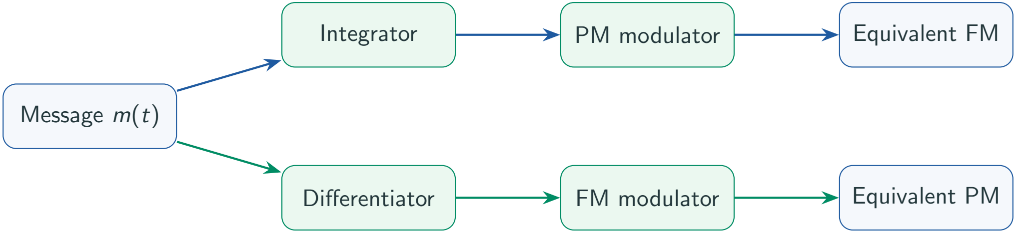
$$s_{\mathrm{FM}}(t) = A_c \cos[2\pi f_c t + \beta_f \sin(2\pi f_m t)]$$

FM uses sine; PM uses cosine

The sine/cosine difference

FM integrates the message, so cosine becomes sine. PM uses the message directly, so cosine stays cosine.

FM and PM Are Linked



What this means conceptually

If you integrate the message before a PM modulator, you get an FM-like result. If you differentiate the message before an FM modulator, you get a PM-like result. The two are mathematically close even though their direct definitions sound different.

Equation Summary — AM Family

Technique	Equation	Main Idea
Standard AM	$s(t) = A_c[1 + k_a m(t)]\cos(2\pi f_c t)$	Envelope follows message; carrier present
DSB-SC	$s(t) = A_c m(t)\cos(2\pi f_c t)$	Carrier suppressed; double sideband
SSB	$s(t) = \text{USB xor LSB}$	Single sideband; half the bandwidth
VSB	$S(f) = S_{\text{DSB}}(f)H_{\text{VSB}}(f)$	One sideband plus vestige filter

Equation Summary — Angle Modulation

Technique	Equation	Main Idea
FM	$s(t) = A_c \cos[2\pi f_c t + 2\pi k_f \int m(t) dt]$	Frequency follows message; constant envelope
PM	$s(t) = A_c \cos[2\pi f_c t + k_p m(t)]$	Phase follows message; constant envelope

AM, FM, and PM Compared

Scheme	Message changes	Usually constant	Main strength	Main weakness
AM	Carrier amplitude	Frequency and phase	Simple receiver and clear carrier intuition	Sensitive to amplitude noise; carrier wastes power
FM	Instantaneous frequency	Amplitude	Good noise performance; constant envelope	Requires more bandwidth
PM	Instantaneous phase	Amplitude	Closely linked to later phase-based digital modulation	Harder to picture intuitively at first

One clean sentence

AM puts the message in the **height** of the carrier. FM and PM put the message in the **timing/angle behavior** of the carrier.

Where the AM Family Appears in Practice

Scheme	Typical use or historical role	Why that use makes sense
AM	Medium-wave broadcast radio; aviation voice AM	Simple receivers; easy envelope detection; legacy compatibility
DSB-SC	Conceptual building block in modulation theory and mixers	Shows that the carrier can be removed if the receiver can recreate it
SSB	Narrowband long-range voice links, amateur radio	Bandwidth is valuable, so sending one sideband is attractive
VSB	Legacy analog television video	Practical compromise when pure SSB filtering is difficult

Where FM and PM Appear in Practice

Scheme	Typical use	Why it fits
FM	VHF broadcast radio, two-way radio	Better resistance to amplitude noise and distortion
PM	Conceptual bridge to phase-based digital modulation	Helps students later understand PSK and related schemes

Why this still matters

Even when later systems are digital, they still inherit the carrier intuition developed here. FM and PM show that **where the message lives in the waveform** changes the receiver design and noise behavior.

Why Learn Analog Modulation in a Mostly Digital World?

1. **It still exists directly.** AM and FM broadcasting are not just historical examples.
2. **It builds carrier intuition.** Digital modulation still changes amplitude, frequency, or phase of a carrier.
3. **It introduces system tradeoffs early.** Bandwidth, power, noise sensitivity, and receiver complexity all appear clearly here.
4. **It explains later topics.** Topic 4 converts analog sources to bits; Topic 5 then modulates those bits onto a carrier.

The big bridge

This topic answers: **how do we move an analog message onto a carrier?** The next two topics answer: **should we digitize that analog source, and if we do, how do we carry the resulting bits efficiently?**

Summary

- Analog modulation exists because many practical channels are band-pass while many messages begin at baseband
- A carrier offers three basic knobs: amplitude, frequency, and phase
- AM makes the envelope follow the message and needs bandwidth $2B$
- DSB-SC, SSB, and VSB are efficiency-oriented variants of the AM idea
- FM and PM are angle-modulation methods that keep amplitude nearly constant
- FM generally spends more bandwidth to gain better resistance to amplitude noise
- The carrier intuition learned here becomes the foundation for Topics 4 and 5

Next: Topic 4 asks when and why we convert an analog source into samples, quantized values, and finally bits.

References

1. B. A. Forouzan, *Data Communications and Networking*, 5th ed., Ch. 5, “Analog Transmission,” especially §5.2, pp. 147–151
2. T. S. Rappaport, *Wireless Communications: Principles and Practice*, 2nd ed., Ch. 6, “Modulation Techniques for Mobile Radio”
   The original short deck pointed to Ch. 5; that was incorrect for modulation. This expanded deck uses Ch. 6. Page numbers in the appendix note the edition nuance explicitly.
3. A. Goldsmith, *Wireless Communications*, Ch. 5, pp. 126–157 for the later bridge to passband and digital modulation ideas

Appendix — First-Exposure Terminology I

Term	Meaning in this topic
Baseband	Original signal region before shifting to a carrier; usually near 0 Hz
Band-pass / passband	A nonzero frequency region centered around some carrier frequency
Carrier	High-frequency sinusoid used as the transport waveform for the message
Modulation	Process of changing a carrier parameter according to the message
Demodulation	Process of recovering the message from the received modulated waveform
Envelope	Outer smooth boundary of an AM waveform
Sideband	Shifted copy of the message spectrum that appears around the carrier frequency
Lower sideband (LSB)	The sideband below the carrier frequency
Upper sideband (USB)	The sideband above the carrier frequency
Modulation index / depth	Measure of how strongly the message changes the carrier

Appendix — First-Exposure Terminology II

Term	Meaning in this topic
Over-modulation	AM case where the envelope no longer cleanly represents the message
Envelope detector	Simple AM receiver block that follows the envelope directly
Coherent detection	Detection that needs a locally generated carrier reference matched in frequency and phase
DSB-SC	Double Sideband Suppressed Carrier; both sidebands sent, carrier removed
SSB	Single Sideband; one sideband transmitted to save bandwidth
VSB	Vestigial Sideband; one full sideband plus a small vestige of the other
Angle modulation	Family in which information is carried by frequency or phase behavior rather than amplitude
Instantaneous frequency	The local, moment-by-moment oscillation rate of the carrier
Frequency deviation	Maximum shift of instantaneous frequency away from the carrier frequency
Limiter / discriminator	Receiver blocks commonly used in FM reception to remove amplitude noise and then detect frequency changes

Appendix — Source Mapping I

Textbook page numbers below refer to the printed page numbering of the cited books.

Slide portion	Source	Source pages used
Baseband to Passband: The Big Picture	Instructor bridge + Topic 2 + v1	Not taken directly from one book page; added to explain why analog modulation belongs here for students without a signals prerequisite
The Carrier Gives Us Three Knobs		
Same Carrier, Three Different Behaviors	Forouzan + Topic 2	Forouzan, Ch. 5 introduction and §5.2, pp. 147–148; also consistent with Topic 2 signal terminology
AM: Let the Envelope Follow the Message		
AM in the Time Domain		
AM Bandwidth and Sidebands	Forouzan	§5.2.1, pp. 147–149

Appendix — Source Mapping II

Slide portion	Source	Source pages used
What Information Is Actually Useful in AM?		
Modulation Depth: Too Little, Just Right, Too Much		
Why Ordinary AM Was Attractive	Forouzan + Rappaport + synthesized	Forouzan, §5.2.1, pp. 147–149; Rappaport, Ch. 6, §6.1–§6.2, especially pp. 255–264 in the currently accessible Cambridge updated second edition; summary wording simplified for first-time learners
From AM to the AM Family		
DSB-SC: Save Carrier Power		
SSB: Save Bandwidth Too		
How the Message Is Recovered	Rappaport + synthesized	Rappaport, Ch. 6, §6.2–§6.2.3, pp. 257–264 in the accessible updated second edition; DSB-SC framing is a conceptual bridge built from the same AM/SSB ideas without Fourier-analysis detail
VSB: A Practical Compromise		
AM Family Compared	Synthesized	

Appendix — Source Mapping III

Slide portion	Source	Source pages used
Angle Modulation: Keep Amplitude Constant		
FM: Let the Message Bend the Frequency		
FM in the Time Domain		
Frequency Deviation and FM Bandwidth		
Why FM Handles Amplitude Noise Better	Forouzan + Rappaport	Forouzan, §5.2.2, pp. 149–150; Rappaport, §6.3–§6.3.4, pp. 264–276 in the accessible updated second edition
PM: Let the Message Bend the Phase		
FM and PM Are Linked		
AM, FM, and PM Compared	Forouzan + Rappaport + synthesized	Forouzan, §5.2.3, pp. 150–151; Rappaport, §6.3, pp. 264–266 in the accessible updated second edition; relational wording simplified to stay within the prerequisite constraints
Where the AM Family Appears in Practice		
Where FM and PM Appear in Practice		
Why Learn Analog Modulation in a Mostly Digital World?		
Summary and terminology appendix	Forouzan + Goldsmith + synthesized	Forouzan, pp. 148–150 for AM/FM broadcasting bandwidth examples; Goldsmith, Ch. 5, pp. 142–157 only for the bridge to later passband/digital-modulation thinking; applications and appendix wording synthesized for this course